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Batch 15

Algorithm

Main Algorithm

1. Start.
2. Declare a character array `str`.
3. Read a string from the user.
4. Display the menu:
 1. Count spaces, digits, and special characters.
 2. Count frequency of a given character.
 3. Toggle the case of each character.
 4. Exit.
5. Read the user's choice.
6. Use `switch`:
 - Choice 1 → call `countCharacters()`.
 - Choice 2 → call `characterFrequency()`.
 - Choice 3 → call `toggleCase()`.
 - Choice 4 → terminate the program.
 - Otherwise → display "Invalid choice!".
7. Repeat the menu until choice 4 is selected.
8. Stop.

Function 1: Count Spaces, Digits, and Special Characters

1. Initialize `spaces`, `digits`, and `special` to 0.
2. Traverse the string character by character.
3. If the character is a space, increment `spaces`.
4. Else if it is between '0' and '9', increment `digits`.
5. Else if it is not an uppercase or lowercase alphabet, increment `special`.
6. Display all three counts.

Function 2: Frequency of a Given Character

1. Read the character to be searched.
2. Initialize `count = 0`.
3. Traverse the string.
4. Whenever the current character equals the given character, increment `count`.
5. Display the frequency.

Function 3: Toggle Case

1. Traverse the string character by character.
2. If a character is between 'a' and 'z', convert it to uppercase.
3. If a character is between 'A' and 'Z', convert it to lowercase.
4. Leave digits, spaces, and special characters unchanged.
5. Display the modified string.

C Program

```
#include <stdio.h>

void countCharacters(char str[]);
void characterFrequency(char str[]);
void toggleCase(char str[]);

int main()
{
    char str[200];
    int choice;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    /* Remove newline added by fgets */
    int i = 0;
    while (str[i] != '\0')
    {
        if (str[i] == '\n')
        {
            str[i] = '\0';
            break;
        }
        i++;
    }

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Count spaces, digits, and special\ncharacters\n");
        printf("2. Count frequency of a given character\n");
        printf("3. Toggle the case of each character\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
    } while (choice != 4);
}
```

```

switch (choice)
{
    case 1:
        countCharacters(str);
        break;

    case 2:
        characterFrequency(str);
        break;

    case 3:
        toggleCase(str);
        break;

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice! Please enter 1 to
4.\n");
}

} while (choice != 4);

return 0;
}

void countCharacters(char str[])
{
    int i = 0;
    int spaces = 0, digits = 0, special = 0;

    while (str[i] != '\0')
    {
        if (str[i] == ' ')
        {
            spaces++;
        }
        else if (str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if (!(str[i] >= 'A' && str[i] <= 'Z') ||
(str[i] >= 'a' && str[i] <= 'z'))

```

```

        {
            special++;
        }

        i++;
    }

    printf("\nNumber of spaces = %d\n", spaces);
    printf("Number of digits = %d\n", digits);
    printf("Number of special characters = %d\n", special);
}

```

```

void characterFrequency(char str[])
{
    char ch;
    int i = 0, count = 0;

    printf("Enter the character to search: ");
    scanf(" %c", &ch);

    while (str[i] != '\0')
    {
        if (str[i] == ch)
        {
            count++;
        }
        i++;
    }

    printf("Frequency of '%c' = %d\n", ch, count);
}

```

```

void toggleCase(char str[])
{
    int i = 0;

    while (str[i] != '\0')
    {
        if (str[i] >= 'a' && str[i] <= 'z')
        {
            str[i] = str[i] - 32;
        }
        else if (str[i] >= 'A' && str[i] <= 'Z')
        {
            str[i] = str[i] + 32;
        }
    }
}

```

```

        }

        i++;
    }

    printf("String after toggling case: %s\n", str);
}

```

Test Case Table

Test	Input String	Choice	Additional Input	Expected Output
1	Hello 123!	1	—	Spaces = 1, Digits = 3, Special = 1
2	C Programming 2026@	1	—	Spaces = 2, Digits = 4, Special = 1
3	banana	2	a	Frequency of a = 3
4	Hello World	2	l	Frequency of l = 3
5	Hello123	2	x	Frequency of x = 0
6	Hello World!	3	—	hELLO wORLD!
7	ABCxyz123@	3	—	abcXYZ123@
8	123 @#\$	3	—	123 @#\$
9	Any string	4	—	Program exits
10	Any string	5	—	Invalid choice! Please enter 1 to 4.

Output

```

Last login: Wed Sep 30 08:30:00 on ttys000
[bhavya@Bhavyas-MacBook-Air ~ % nano day38.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day38.c -o day38
[bhavya@Bhavyas-MacBook-Air ~ % ./day38
Enter a string: Hello 123!

----- MENU -----
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 1

Number of spaces = 1
Number of digits = 3
Number of special characters = 1

----- MENU -----
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 2
Enter the character to search: 1
Frequency of '1' = 1

----- MENU -----
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 3
String after toggling case: hELLO 123!

----- MENU -----
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: 3
String after toggling case: Hello 123!

----- MENU -----
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle the case of each character
4. Exit
Enter your choice: █

```